Grade 9 Geography – Study Notes

These notes follow the standard Grade 9 CAPS content. Check them against your school's exam guidelines.

Map skills

Types of maps

Map	Description
Topographic map	Shows natural and man-made features with symbols and contour lines. South African topographic maps are **1:50 000**
Orthophoto map	An aerial photograph with contour lines and names added; scale **1:10 000** (shows more detail of a smaller area)
Atlas maps	Small-scale maps of large areas (countries, the world)
GIS (Geographic Information System)	A computer system that captures, stores, analyses and displays spatial (location) data in **layers** (e.g. roads, rivers, schools)

Scale

Scale shows the relationship between a distance on the map and the real distance on the ground.

Type	Example
Word scale	"1 cm represents 0.5 km"
Ratio (numerical) scale	1 : 50 000 (1 cm on the map = 50 000 cm on the ground = 500 m)
Line (linear) scale	A ruler-like bar divided into km

- **Large scale** (e.g. 1:10 000) = small area, lots of detail. **Small scale** (e.g. 1:1 000 000) = large area, little detail.
- **Calculating real distance:** measure on the map in cm, then multiply by the scale. On a 1:50 000 map, **1 cm = 0.5 km**, so 6 cm = 6 × 0.5 = **3 km**. On a 1:10 000 orthophoto map, **1 cm = 100 m (0.1 km)**.
- For curved distances (roads, rivers) use a piece of string or the edge of a paper.

Location: grid references (latitude and longitude)

- **Latitude** lines run east–west, measured **north or south of the Equator** (0° to 90°).
- **Longitude** lines run north–south, measured **east or west of the Prime (Greenwich) Meridian** (0° to 180°).
- South Africa is **south** of the Equator and **east** of Greenwich, so coordinates look like **33°45′S; 18°58′E** (Paarl area).
- 1 degree (°) = 60 minutes (′); 1 minute = 60 seconds (″). **Always give latitude first.**

Direction and bearing

- **16 compass directions:** N, NNE, NE, ENE, E, ESE, SE, SSE, S, SSW, SW, WSW, W, WNW, NW, NNW.

- Direction is always given **from** one place **to** another ("B is NE of A").
- **True bearing** = the angle measured **clockwise from true north** (0°–360°). East = 90°, South = 180°, West = 270°.
- **Magnetic bearing** = bearing measured from **magnetic north** (where a compass points). Magnetic declination is the angle between true north and magnetic north; it changes every year.

Contour lines and landforms

Contour lines are brown lines joining places of the **same height above sea level**. The **contour interval** on 1:50 000 maps is **20 m**.

Contour pattern	Landform / slope
Lines **close together**	**Steep** slope
Lines **far apart**	**Gentle** slope
Evenly spaced	**Even** (uniform) slope
Close at the top, far apart at the bottom	**Concave** slope
Far apart at the top, close at the bottom	**Convex** slope
Closed circles, highest in the middle	**Hill / conical hill**
Closed shape with a flat top, steep sides	**Mesa** (wide) / **butte** (narrow)
V-shapes pointing **up** to higher ground	**Valley** (river flows in it)
V-shapes pointing **down** to lower ground	**Spur**
Low area between two hills	**Saddle / nek**

- **Spot height** = a dot with the exact height. **Trigonometrical station** = a triangle with the height (survey point).
- **Gradient** = vertical interval (VI) ÷ horizontal equivalent (HE), written as 1 : x. VI = height difference; HE = real distance (same units!). *Example:* VI = 100 m, HE = 2 000 m → gradient = 100 ÷ 2 000 = **1 : 20**.
- **Cross-section:** a side view of the landscape along a line between two points, drawn from the contour heights.
- **Intervisibility:** two points can see each other if no higher ground lies between them (check on a cross-section).

Map symbols and conventional signs

Learn the key (legend): roads (national, main, secondary), railways, rivers (perennial = flows all year; non-perennial = dry part of the year), dams, buildings, schools, churches, power lines, cultivated land, orchards, vegetation, cemeteries, airfields.

Development issues

Development is the process of improving people's **quality of life** — not only income, but also health, education, housing, safety and freedom.

Measuring development (indicators)

Indicator	What it measures	Developed country	Developing country
GDP / GNI per capita	Average income per person per year	High	Low
Life expectancy	Average number of years a person is expected to live	High (80+)	Lower (50–65)
Literacy rate	% of adults who can read and write	Near 100%	Lower
Infant mortality rate	Babies who die before age 1, per 1 000 births	Low	High
Population growth rate	How fast the population grows	Slow	Fast
Access to clean water and sanitation	% of people with safe water and toilets	Almost all	Fewer
HDI (Human Development Index)	Combines income, education and life expectancy; a value from 0 to 1	Close to 1	Lower

GDP = total value of goods and services produced in a country in a year. **Per capita** = per person.

- **Economic indicators** measure wealth (GDP, income). **Social indicators** measure quality of life (health, education, housing).
- One indicator alone is not enough: an average hides **inequality** (South Africa has one of the highest levels of inequality in the world).

Developed and developing countries

- **Developed** ("economically more developed", the **North**): USA, UK, Germany, Japan, Australia.
- **Developing** ("less developed", the **South**): most of Africa, parts of Asia and South America.
- **Emerging economies** (developing fast): South Africa, Brazil, India, China (with Russia these form **BRICS**).

Factors that affect development

Factor	Effect
History (colonialism, apartheid)	Resources were taken; many people were denied education and land
Trade	Developing countries export cheap **raw materials** and import expensive **manufactured goods** (unfair trade)
Debt	Money spent on repaying loans cannot be spent on schools and hospitals
Corruption and poor governance	Money meant for development is stolen or wasted
Conflict and war	Destroys infrastructure and stops development
Natural disasters, climate, disease	Droughts, floods, HIV/AIDS and malaria slow development
Education and skills	An educated workforce attracts investment and earns more
Infrastructure	Roads, electricity, water and communication are needed for businesses
Population growth	Fast growth puts pressure on services and jobs

Trade

- **Exports** = goods sold to other countries; **imports** = goods bought from other countries.
- **Balance of trade** = exports − imports. A **surplus** if exports are more; a **deficit** if imports are more.
- **Fair trade** = producers in developing countries get a fair price and decent working conditions.
- **Trade barriers:** tariffs (taxes on imports), quotas (limits on amounts).
- **Adding value** (processing raw materials before exporting, e.g. turning cocoa into chocolate) earns more money and creates jobs.

Development aid

Type	Example	Advantage / disadvantage
Emergency (short-term) aid	Food, tents and medicine after a disaster	Saves lives but doesn't solve long-term problems
Long-term (development) aid	Building schools, clinics, wells; training	Helps people become self-sufficient
Bilateral aid	One government to another	May come with **conditions** ("tied aid")
Multilateral aid	Through organisations such as the UN, World Bank	Larger scale
NGO aid	Red Cross, Oxfam, Gift of the Givers	Works directly with communities

Aid can create dependency if it replaces local production. The best aid helps communities help themselves (e.g. teaching skills, small business loans).

Sustainable development

Sustainable development meets the needs of people today **without harming the ability of future generations** to meet their needs. It balances **economic growth, social needs and the environment**. Examples: renewable energy, recycling, community food gardens, eco-tourism, protecting water sources.

Resource use and sustainability

A **natural resource** is anything from nature that people use: water, soil, air, minerals, forests, fish, wildlife, sunlight.

Types of resources

Type	Meaning	Examples
Renewable	Replaced naturally if used wisely	Water, soil, forests, fish, sun, wind
Non-renewable	Finite; takes millions of years to form, will run out	Coal, oil, natural gas, gold, platinum, iron ore
Flow resources	Never run out regardless of use	Sunlight, wind, tides

Over-use of renewable resources can make them non-renewable in practice (e.g. over-fishing, deforestation, soil erosion).

Water

South Africa is a **water-scarce** country: low and unreliable rainfall (average about 450 mm per year, below the world average), high evaporation, frequent droughts.

- **Uses:** agriculture (irrigation uses the most), domestic, industry, mining, power stations.
- **Problems:** droughts (e.g. Cape Town's "Day Zero" 2018), pollution (sewage, mining acid water, fertilisers), leaking pipes, unequal access.
- **Water management:** dams and inter-basin transfer schemes (e.g. **Lesotho Highlands Water Project** supplies Gauteng), water restrictions, fixing leaks, rainwater tanks, recycling grey water, drip irrigation, protecting wetlands.

Soil

Soil takes hundreds of years to form but can be lost in one storm.

- **Soil erosion** = removal of topsoil by water or wind.
- **Causes:** overgrazing, deforestation, removing vegetation, ploughing up and down slopes, poor farming, veld fires.
- **Effects:** less fertile land, lower crop yields, food shortages, dongas (gullies), silting of dams and rivers.
- **Soil conservation:** contour ploughing (along the contours), terracing on slopes, planting trees and cover crops, crop rotation, rotational grazing, stopping overgrazing, building small walls (gabions) in dongas.

Forests and biodiversity

- **Deforestation** = clearing forests for farming, timber, firewood or building.
- **Effects:** loss of habitats and **biodiversity** (the variety of plant and animal life), soil erosion, less rainfall, more CO_2 (climate change).
- **Solutions:** replanting (reforestation), sustainable logging, protected areas (national parks, nature reserves), using alternatives to firewood.

Minerals and energy

- South Africa is rich in minerals: **gold, platinum, coal, diamonds, iron ore, manganese, chrome**. Mining creates jobs and export income.
- **Problems of mining:** non-renewable, scars the landscape, mine dumps, **acid mine drainage** polluting water, air pollution, dangerous working conditions.
- **Coal** provides most of SA's electricity but causes air pollution and climate change. Alternatives: solar, wind, hydro.

Pollution

Type	Causes	Solutions
Air	Factories, power stations, vehicles, burning	Filters, cleaner fuels, public transport, renewable energy
Water	Sewage, factory waste, fertilisers, litter, mining	Treat waste water, laws, clean-ups
Land	Litter, illegal dumping, landfills, mine dumps	Reduce, reuse, recycle; proper waste removal

Sustainable use of resources

Sustainable use = using resources in a way that meets present needs without destroying them for the future.

- **The 3 Rs:** Reduce, Reuse, Recycle.
- Use renewable energy; conserve water and electricity.
- Protect ecosystems (wetlands, forests, marine protected areas); fishing quotas.
- Laws and organisations: Department of Forestry, Fisheries and the Environment; SANParks; WWF.
- **Conservation** = protecting and managing resources carefully; **preservation** = keeping something untouched.

Surface forces that shape the Earth

The Earth's surface is shaped by **internal forces** (from inside: volcanoes, earthquakes, folding, faulting) and **external (surface) forces** that break down and move rock: **weathering, erosion and deposition**.

Weathering

Weathering = the breaking down of rock **in place** (it does not move).

Type	How it works	Examples	Climate
Physical (mechanical)	Rock breaks into smaller pieces without changing its chemical make-up	**Frost action** (water freezes in cracks, expands, splits rock); **exfoliation** (heating and cooling cause outer layers to peel off like an onion); plant roots growing in cracks	Dry areas with big temperature ranges; cold mountains
Chemical	Minerals in the rock change or dissolve	**Carbonation** (rainwater with CO_2 dissolves limestone → caves); **oxidation** (iron rusts, rock turns red); **hydrolysis**	Hot, wet areas
Biological	Living things break rock	Roots, burrowing animals, lichens producing acids	All climates

Erosion, transport and deposition

- **Erosion** = the wearing away **and removal** of rock and soil by agents: **water (rivers), wind, ice (glaciers), waves (sea) and gravity**.
- **Transport** = moving the eroded material.
- **Deposition** = dropping the material when the agent loses energy (slows down) — forming new landforms.
- **Mass movement** = downslope movement of material due to gravity: landslides, mudflows, rockfalls, soil creep.

Rivers and river landforms

Stage	Features	Landforms
Upper course (source, mountains)	Steep gradient, fast flow, **vertical erosion** (cuts down)	**V-shaped valleys**, waterfalls, rapids, interlocking spurs
Middle course	Gentler gradient, **lateral erosion** (sideways)	Wider valleys, **meanders** (bends)

Stage	Features	Landforms
Lower course (to the mouth)	Very gentle gradient, slow flow, **deposition**	**Flood plains**, **oxbow lakes** (cut-off meanders), **levees**, **deltas** and estuaries at the mouth

- A meander has fast water and erosion on the **outside** bend (river cliff) and slow water and deposition on the **inside** bend (slip-off slope / point bar).
- **Drainage basin** (catchment area) = the area drained by a river and its tributaries; **watershed** = the high ground separating two basins; **tributary** = a smaller river joining the main river; **confluence** = where two rivers meet.

Wind (aeolian) landforms

In deserts and dry areas: **sand dunes** (deposition), **mushroom rocks** / pedestal rocks (wind erosion at the base), and **deflation hollows** (wind removes loose material).

Coastal landforms

Erosion (destructive waves)	Deposition (constructive waves)
Cliffs and **wave-cut platforms**	**Beaches**
Caves → arches → stacks → stumps (a headland is eroded in this order)	**Spits** (sand extending into the sea)
Headlands and bays (hard rock forms headlands; soft rock erodes into bays)	**Sandbars** and lagoons

Human impact and hazards

- People speed up erosion through **deforestation, overgrazing, mining, building on slopes** and poor farming.
- Hazards: **floods** (e.g. KwaZulu-Natal 2022), **landslides**, coastal erosion damaging property.
- Reduce the impact: plant vegetation on slopes, build retaining walls and gabions, avoid building on flood plains, early-warning systems.